Module 2 – Prompt Engineering Essentials

Demo 2 - Designing Creative Prompts for Diverse NLP Tasks

Objective

To demonstrate how to design creative prompts for various NLP tasks. Learners will explore how to craft effective prompts for each task, ensuring high-quality outputs from large language models (LLMs).

Problem Statement : Automating NLP Solutions at InnovateTech

InnovateTech, a company specializing in AI-powered automation solutions, there is a constant need to automate a wide variety of text-based tasks across different industries. These tasks include summarizing documents, generating code, answering complex queries, translating text between languages, classifying documents, and extracting key information from data. The company needs to generate consistent, high-quality outputs for these diverse NLP tasks using LLMs.

Key Challenge

The challenge lies in crafting effective prompts for each specific NLP task. Since these tasks vary greatly in nature, the company must design different prompt strategies that guide the model to produce relevant and accurate responses for each use case.

InnovateTech's Target Deliverables

At **InnovateTech**, the following NLP tasks need to be efficiently handled using LLMs:

- **Text Summarization**: Summarizing long documents, articles, or reports to provide concise, informative versions.
- **Code Generation**: Generating functional code based on natural language descriptions of software functionality.
- **Question Answering**: Responding to factual or open-ended questions based on provided documents or databases.
- **Machine Translation**: Translating text from one language to another while preserving meaning and context.

- **Text Classification**: Categorizing text documents into predefined categories or labels.
- **Data Extraction**: Extracting specific information or entities from unstructured text or datasets.

Approach

To address this challenge, we can use creative prompting strategies for each NLP task. The tasks will be handled sequentially to allow the model to generate the most accurate and relevant output for each case.

Solution

Note: The following prompt solutions are delivered by ChatGPT, an AI language model developed by OpenAI.

Text Summarization

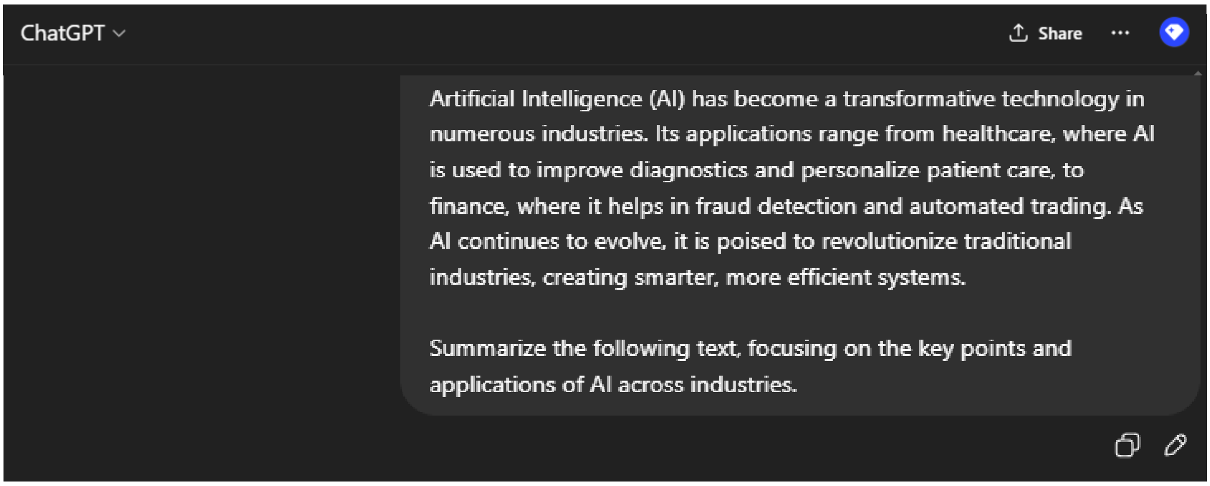
In text summarization, the model is tasked with condensing long pieces of text into concise summaries. This is ideal for quickly extracting key points or providing high-level overviews.

Prompt:

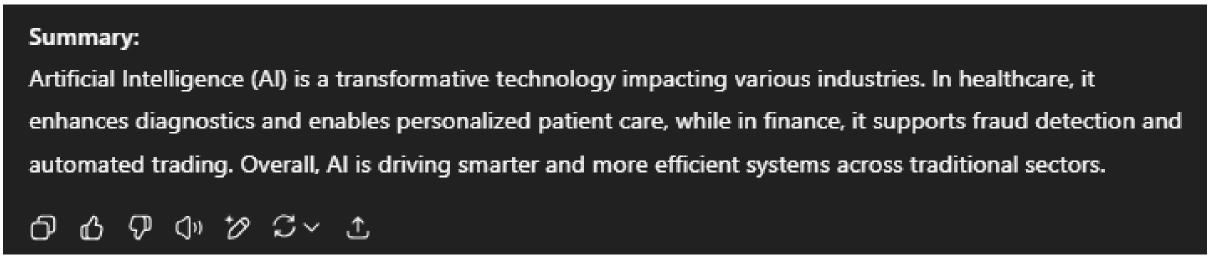
"Input Text:

Artificial Intelligence (AI) has become a transformative technology in numerous industries. Its applications range from healthcare, where AI is used to improve diagnostics and personalize patient care, to finance, where it helps in fraud detection and automated trading. As AI continues to evolve, it is poised to revolutionize traditional industries, creating smarter, more efficient systems.

Summarize the following text, focusing on the key points and applications of AI across industries."



ChatGPT Response:

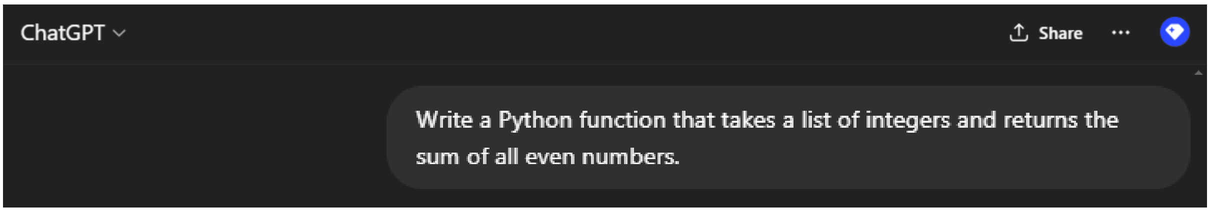


By using this prompt, **InnovateTech** was able to automate the summarization of extensive research papers, significantly reducing the time needed to extract essential information for stakeholders. This allowed the team to quickly disseminate important research findings across departments without losing critical details.

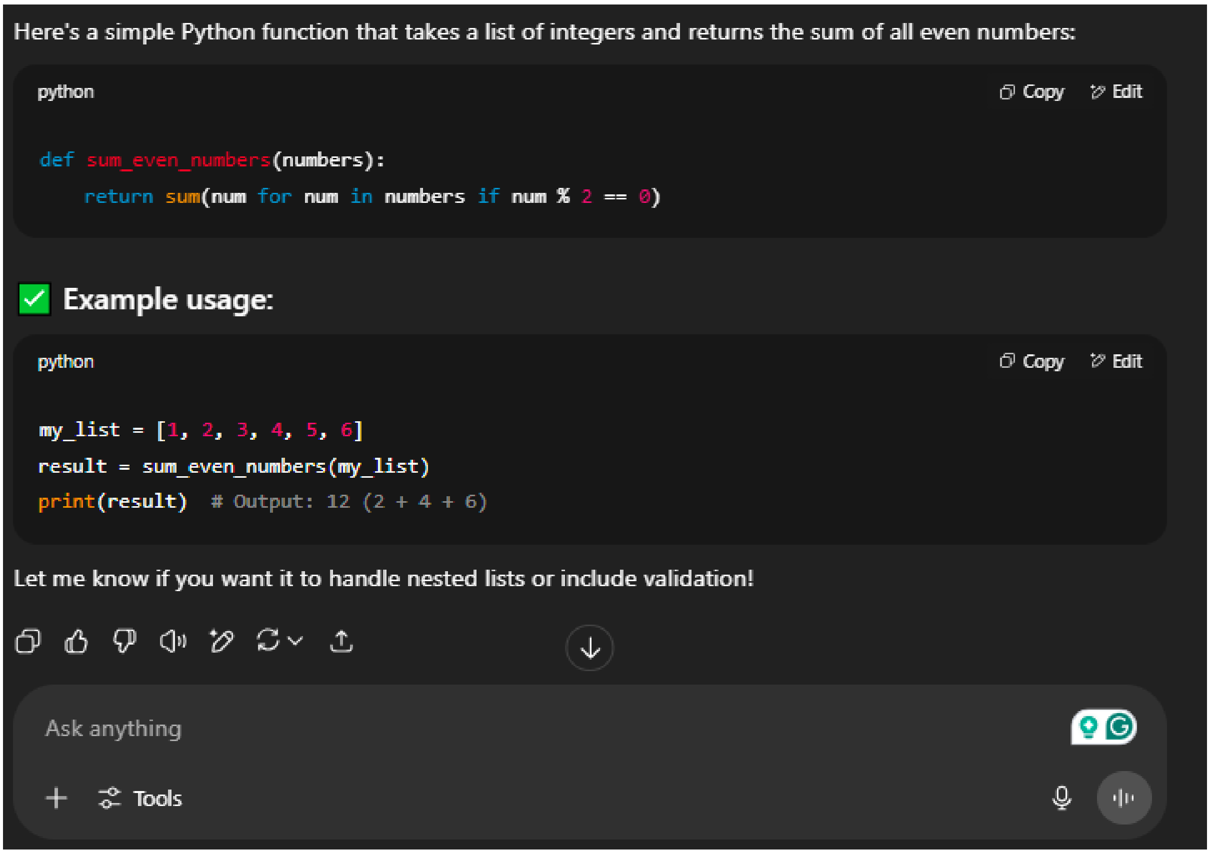
Code Generation

In **code generation**, the model is guided to produce working code from natural language descriptions, a useful tool for automating repetitive coding tasks.

Prompt:
"Write a Python function that takes a list of integers and returns the sum of all even numbers."



ChatGPT Response:



After implementing this prompt, **InnovateTech** successfully automated the generation of boilerplate code for common functionalities. This allowed the development team to focus more on complex tasks while the system handled routine code generation, reducing errors and improving efficiency.

Question Answering

Question answering involves providing answers based on a given set of information or context. This task can range from factual to opinion-based questions.

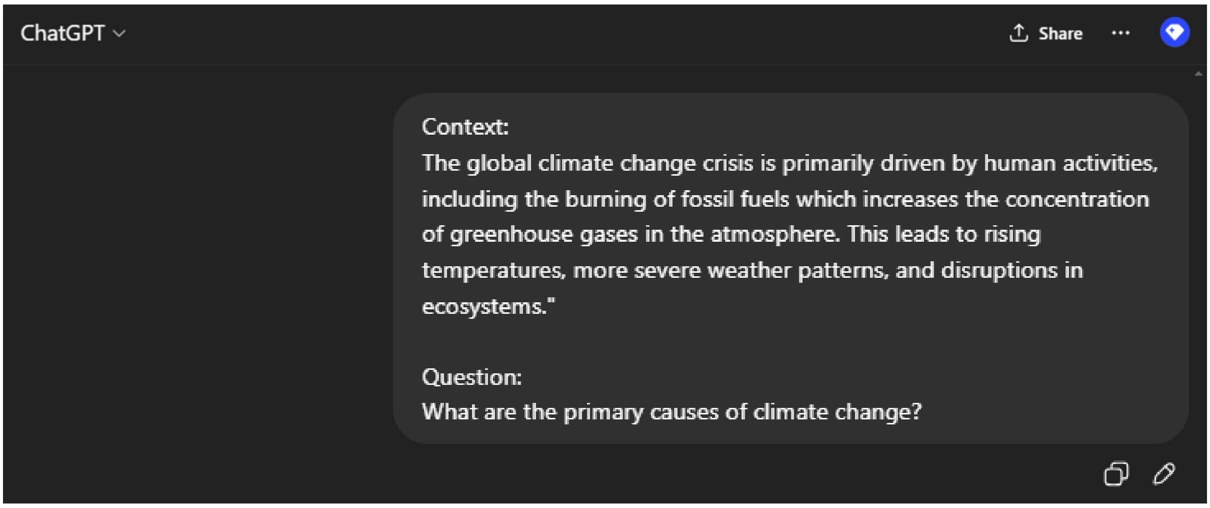
Prompt:

“Context:

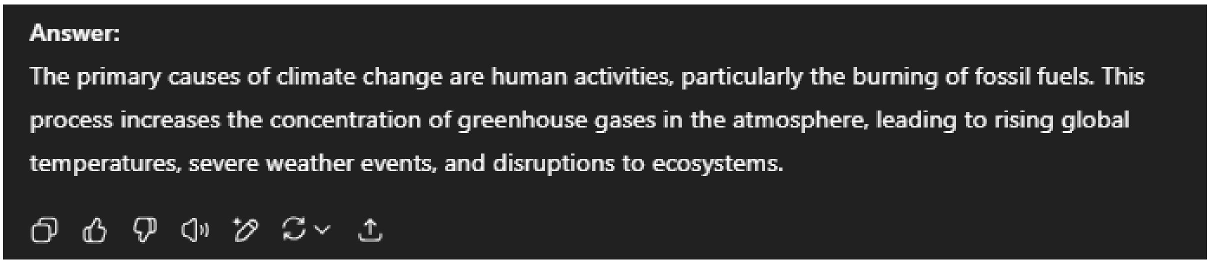
The global climate change crisis is primarily driven by human activities, including the burning of fossil fuels which increases the concentration of greenhouse gases in the atmosphere. This leads to rising temperatures, more severe weather patterns, and disruptions in ecosystems."

Question:

What are the primary causes of climate change?"



ChatGPT Response:



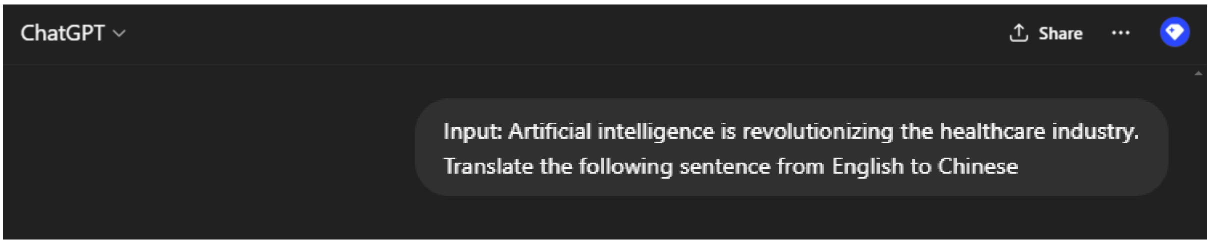
InnovateTech integrated question answering into their customer support system, allowing the model to automatically answer common queries about climate change based on articles, reports, and other documents. This resulted in faster response times for customers and a reduction in manual research efforts.

Machine Translation

In machine translation, the model translates text from one language to another while maintaining meaning and context.

Prompt:
"Input: Artificial intelligence is revolutionizing the healthcare industry.

Translate the following sentence from English to Chinese."



ChatGPT Response:

Using this translation prompt, **InnovateTech** was able to launch a multi-lingual documentation platform, making its technical materials accessible to a global audience. The model accurately translated documents, ensuring consistency and preserving technical accuracy across languages.

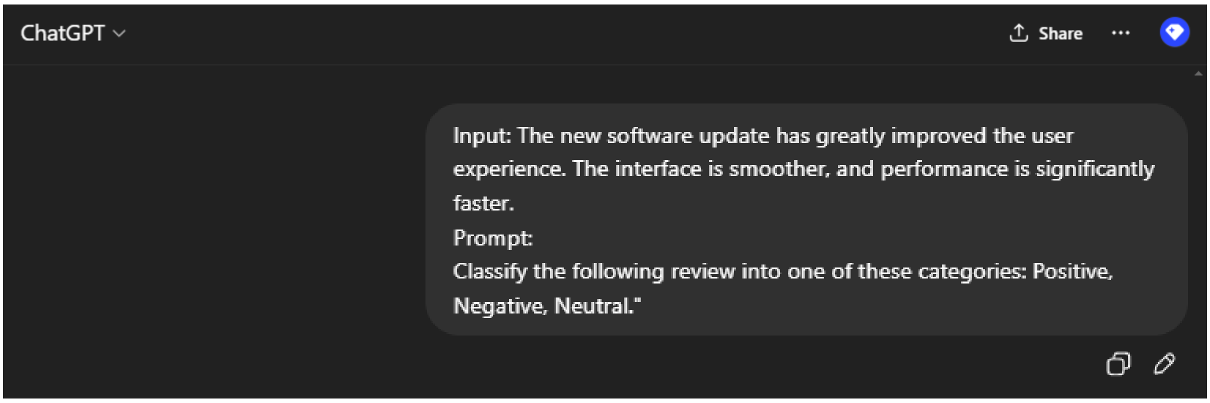
Text Classification

In text classification, the model is asked to categorize text into predefined categories based on its content. This is useful for organizing large volumes of unstructured text.

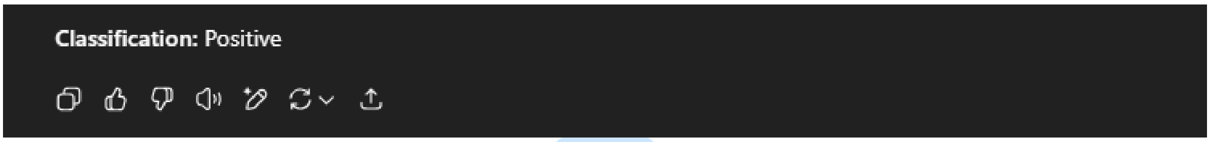
Prompt:
" Input: The new software update has greatly improved the user experience. The interface is smoother, and performance is significantly faster.

Prompt:

Classify the following review into one of these categories: Positive, Negative, Neutral."



ChatGPT Response:



InnovateTech implemented this text classification model to automatically categorize user feedback into sentiment categories. This allowed the company to prioritize responses based on customer sentiment and gain valuable insights into product performance, streamlining the review analysis process.

Data Extraction

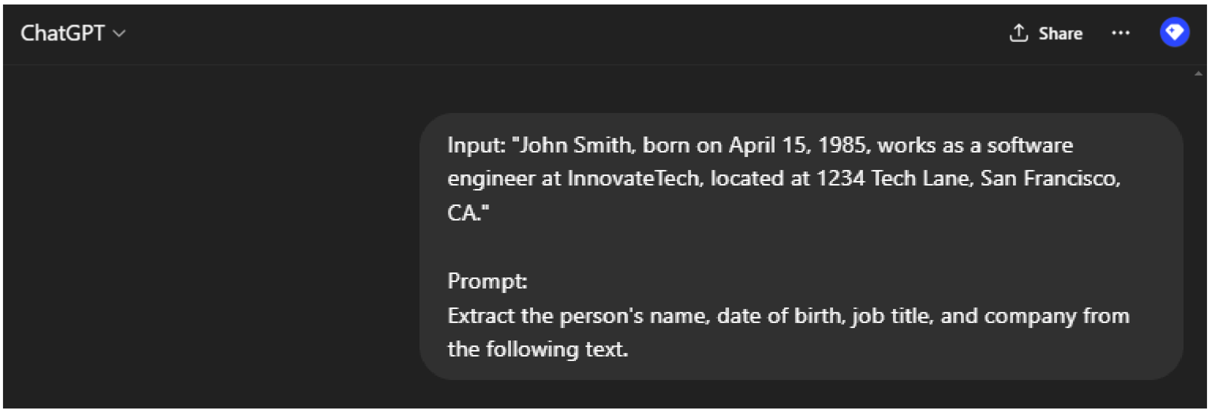
In data extraction, the model is tasked with extracting specific pieces of information from a larger dataset or text.

Prompt:

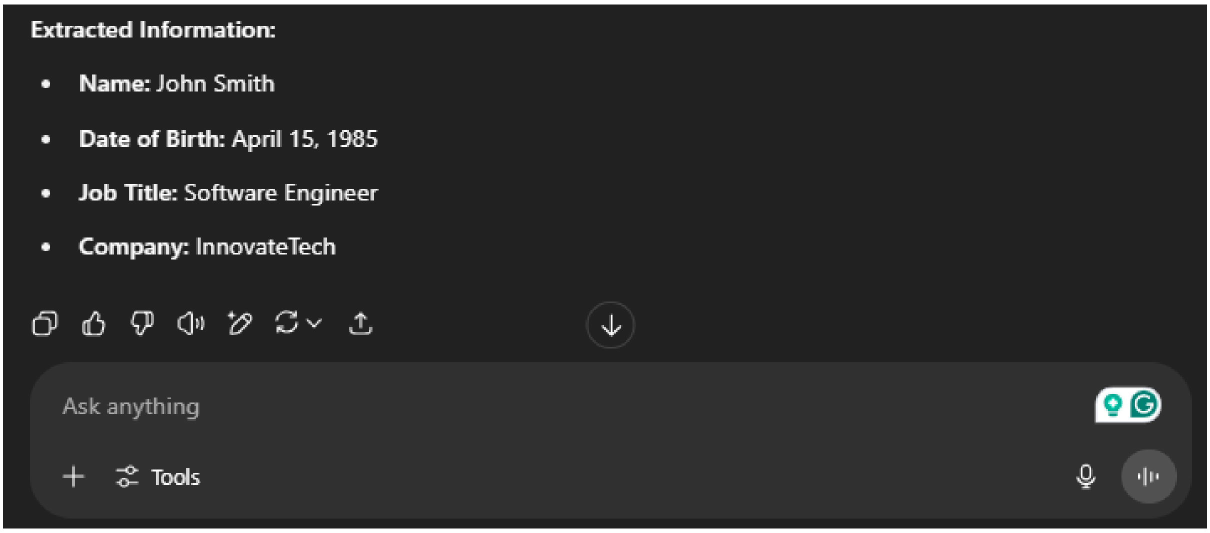
“Input: "John Smith, born on April 15, 1985, works as a software engineer at InnovateTech, located at 1234 Tech Lane, San Francisco, CA."

Prompt:

Extract the person's name, date of birth, job title, and company from the following text."



ChatGPT Response:



By using this data extraction prompt, **InnovateTech** developed an automated system to extract key dates from legal documents, contracts, and reports. This saved hours of manual data entry and ensured that the information was accurate and readily available for further analysis.

Conclusion

By designing creative prompts for each NLP task, **InnovateTech** was able to efficiently generate high-quality, contextually relevant content tailored to a variety of business needs. The tasks included:

- **Text Summarization** helped quickly extract key points from long documents.

- **Code Generation** automated the creation of functional code, allowing developers to focus on more complex tasks.
- **Question Answering** improved customer support by providing quick answers to common queries.
- **Machine Translation** facilitated global accessibility by translating technical documents across languages.
- **Text Classification** allowed the company to categorize large volumes of text based on sentiment and relevance.
- **Data Extraction** streamlined the process of pulling key information from legal and business documents.

These techniques empowered professionals at **InnovateTech** to automate and streamline a wide range of text-based tasks, saving time and improving overall productivity.